Maya M. Lassiter

I make electronics differently. Most recently, this has been as a Process Engineer at Atomic Semi. My career has built novel electronics and solved field-based engineering challenges. Combining experimental nanofabrication, scalable semiconductor process development, and iterative design for custom manufacture.

Relevant Experience

Senior Process Engineer, Atomic Semi *(1+ years, current position)*
- Led process documentation, benchmarking and analysis for direct write mask making and lithography
- Established scalable doping CMOS on SOI using custom hardware and insights from closed loop testing
- Process Engineer #4/10+, Employee #40/100+

Doctoral Research, Autonomous Microscopic Robots *(6 years)*
- Principal Investigator for the design and fabrication of the world's first autonomous microscopic robot
- Lead experiementalist, responsible for design of experiments, device characterization, testing, and analysis
- Invented foundry-compatible micromachining process at 55- and 28-nm CMOS technology nodes for scalable production
- Deomcratized new technology via open source low-cost kits for using and controlling microscopic robots
- Added to the Museum of Science and Industry and Computer History Museum permanent collections

Nanofabrication Maintenence Technician *(2 years)*
- Directed process documentation and training of new lab users
- Ensured lab upkeep for day to day operations supporting over 200 users
- Support for cleanroom processes across deposition, etching, lithography, packaging, and metrology
- Supported capital equipment move and install during cleanroom relocation

Packaging Engineer, Flexible Biophotonic Neural Probes *(2 years)*
- Developed heterogenous packaging process for chronic optigenetic brain stimulation with soft neural probes
- Invented a low-cost solution with commercial components and custom 3D printed parts
- Miniturized from pumped benchtop laser to <1cm^3 mobile package for untethered chronic implantation

Carnegie Mellon Solar Racing *(2 years)*
- Fabricated custom carbon fiber solar powered boat for international competition
- Built and maintained custom electrical system on-site in The Netherlands
- Pilot for part of multi-day racing through The Netherlands and exhibition in Monaco

Skills

Nanofabrication process development
Chip-scale packaging
Chip validation and mask layout
Test, characterization, and analysis
Python/C

Recognition

University of Pennsylvania Presidential Fellow
MIT Lincoln Lab GEM PhD Fellow
Carnegie Mellon University Outstanding Woman in Engineering
William J. Happel Fellow, GEM MS University Fellow
Carnegie Mellon University Leadership Search Committee Member
Member of: Fontaine Society, IEEE-HKN, SfN, OSA, APS

Education

PhD Electrical and Systems Engineering, University of Pennsylvania
MS Electrical and Computer Engineering, Carnegie Mellon University
BS Electrical and Computer Engineering, Minor in Global Engineering, Carnegie Mellon University